

Estimating Heterogeneous Effects of Tirzepatide vs. Liraglutide on Systolic Blood Pressure Using a Causal Forest

1 Problem And Dataset

1.1 Background

Nearly half of all adults in the United States, approximately 120 million people, have hypertension [1]. Despite this prevalence, hypertension remains poorly controlled, with only one in four hypertensive adults having their blood pressure managed below 130/80 mmHg and approximately 664,000 deaths in 2023 being attributed to the condition [1, 2]. Incretin-based therapies have shown promise in clinical trials for reducing systolic blood pressure (SBP). In the SURMOUNT-1 trial, for example, patients on 15 mg of tirzepatide, a glucagon-like peptide-1 (GLP-1) receptor agonist, weekly saw an estimated placebo-adjusted reduction in SBP of 6.4 mmHg at 72 weeks [3]. While this average reduction statistic is useful for characterizing the overall effect of tirzepatide in the population, a more clinically relevant question is who benefits most in terms of SBP reduction. Thus, using an active-comparator design comparing tirzepatide to liraglutide, an older and less potent GLP-1 receptor agonist, this paper utilizes causal forests to identify if there are subgroups that experience the more potent SBP reductions with tirzepatide over liraglutide.

1.2 Dataset

For this project, the dataset used is a modified version of a larger dataset our lab created to assess the relationship between GLP-1 ingredients and hypertension. This dataset includes first-time users of the GLP-1 ingredients tirzepatide ($n = 606$) and liraglutide ($n = 681$) in the UC Health Data Warehouse with a recorded dispensing prior to September 1, 2024. For our models, the exposure is the index ingredient, tirzepatide or liraglutide, and the outcome is the one-year change in SBP, assessed by taking the difference between follow-up and baseline SBP. Baseline pre-treatment covariates included age, sex, race, ethnicity, baseline BMI, baseline SBP, number of unique antihypertensive medication classes, smoking status, and the presence of the comorbidities: type 2 diabetes, chronic kidney disease, coronary artery disease, and heart failure. Baseline characteristics are summarized in Table 1.

2 The Algorithm

2.1 Why Causal Forests?

Although the average difference in one-year SBP change between tirzepatide and liraglutide users is informative, relying on such a broad summary statistic can obscure clinically meaningful heterogeneity in treatment response. We might not expect, for example, that the effect of tirzepatide versus liraglutide on SBP would be the same among individuals who are overweight as among those with class III obesity. Traditional ways to assess potential heterogeneity include pre-specifying a subgroup and/or assessing interaction terms between relevant covariates. However, the former requires deciding in advance what subgroups are of interest and can inflate the false-positive rate as more subgroups are tested, while the latter requires correctly specifying the functional form of the interaction.

On the other hand, causal forests identify treatment heterogeneity directly from the data, without requiring that subgroups be pre-specified or that the functional relationship be identified. However, in order for the algorithm to perform well, causal forests require a sample large enough to

support decision tree splitting, sufficient overlap between treatment arms, and that all potential confounders are identified and measured.

2.2 What Assumptions Are Necessary For Causal Forests?

In addition to the above performance requirements, causal forests require the following assumptions to hold so that treatment groups are sufficiently comparable, allowing any differences in the outcome to be attributed to variations in the treatment effects [4]:

1. **Consistency:**

$$Y_i = Y_i(W_i) \quad (1)$$

The observed outcome Y_i is the one under the treatment (W_i) actually received.

2. **Overlap:**

$$0 < e(\mathbf{X}_i) < 1 \quad \text{where} \quad e(\mathbf{x}) = \Pr(W_i = 1 \mid \mathbf{X}_i = \mathbf{x}) \quad (2)$$

Every patient could plausibly have received either treatment option.

3. **Unconfoundedness:**

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp W_i \mid \mathbf{X}_i \quad (3)$$

Given covariates $\mathbf{X}_i$, treatment assignment is effectively random.

2.3 How Does The Algorithm Work?

The goal of causal forests is to estimate the conditional average treatment effect (CATE) for every covariate profile in a dataset. Formally, this is expressed as:

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathbf{X}_i = \mathbf{x}] \quad \text{where} \quad \mathbf{x} \in \mathcal{X} \quad (4)$$

where $Y_i(1)$ and $Y_i(0)$ can be thought of as the potential outcomes under tirzepatide and liraglutide, respectively, for a patient with covariate profile $\mathbf{X}_i = \mathbf{x}$. As it is impossible to observe both outcomes for each patient, this individual treatment effect cannot be directly calculated. Instead, causal forests estimate this average treatment effect based on patients with similar covariate profiles.

Causal forests do this task by first "growing" many causal trees, each built off a subsample of the complete dataset. A causal tree can be thought of as a regression tree that estimates an average treatment effect in each leaf

$$\hat{\tau}_\ell = \bar{Y}_\ell^{\text{tirz}} - \bar{Y}_\ell^{\text{lira}} \quad (5)$$

instead of a mean and chooses splits to maximize the variance in the estimated treatment effects across leaves while also penalizing splits based on limited data. This splitting criterion is formalized by [5]:

$$\widehat{\text{EMSE}}_\tau(\mathcal{S}^{\text{tr}}, \Pi) = \underbrace{\frac{1}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{tr}}} \hat{\tau}^2(\mathbf{X}_i; \mathcal{S}^{\text{tr}}, \Pi)}_{\text{Reward: Heterogeneity in } \hat{\tau}} - \underbrace{\frac{2}{N^{\text{tr}}} \sum_{\ell \in \Pi} \left(\frac{S_{\text{tirz}}^2(\ell)}{p} + \frac{S_{\text{lira}}^2(\ell)}{1-p} \right)}_{\text{Penalty: Within-leaf Variance}}, \quad (6)$$

where the first term rewards between-leaf heterogeneity in $\hat{\tau}$ and the second term penalizes within-leaf outcome variance ($S_{\text{tirz}}^2(\ell)$, $S_{\text{lira}}^2(\ell)$) in small leaves. We also clarify that p represents the treated share, N^{tr} the training sample size, $\mathcal{S}^{\text{tr}}$ the training subsample, and Π the partition. It should be

noted that causal trees implement "honest" estimation, which means that prior to the construction of a tree, each tree's sample is split 50:50, with half of the sample then being used to construct the trees and the other half being used to estimate the within-leaf effects.

After constructing many honest causal trees, they are averaged together to produce the CATE estimates that are asymptotically Gaussian with variance estimates derived through the infinitesimal jackknife procedure [4]. This provides us with an estimated effect and 95% confidence interval, $\hat{\tau}(\mathbf{x}) \pm 1.96\sigma_n(\mathbf{x})$, for each covariate profile.

3 Results And Discussion

3.1 Results

For this project, we fit a causal forest using 5,000 causal trees, with tirzepatide as the index ingredient and liraglutide as the comparator. The unadjusted mean one-year SBP changes were -7.9 mmHg and -2.4 mmHg for tirzepatide and liraglutide, respectively. After adjusting for confounding, we estimate the average treatment difference to be -4.82 mmHg (95% CI: -6.28 to -3.37), suggesting that tirzepatide use is associated with a statistically significant reduction in SBP over liraglutide at one year.

To assess our causal forest, we first confirmed sufficient overlap in the propensity scores by graphing the propensity histogram (Figure 1). We then assessed the forest's goodness of fit by evaluating its mean (Est: 1.00; 95% CI: 0.70 to 1.31) and differential (Est: -0.07; 95% CI: -1.31 to 1.16) forest prediction. The former result indicates the model can capture the average effect well, but the latter result suggests that there is no evidence of treatment-effect heterogeneity. We then proceeded to verify that there was no evidence of treatment heterogeneity by assessing the forest's best linear projection and the rank-weighted average treatment effect (RATE). The best linear projection assesses how treatment effects vary with covariates in a regression-like format, and we found no significant effect modifiers, with only baseline SBP ($p = 0.09$) and coronary artery disease ($p = 0.09$) showing borderline signals (Table 2). The RATE, which evaluates whether the forest can identify patients with larger treatment benefits, was also not significant (Est: 0.19; 95% CI: -1.30 to 1.69). Finally, we assessed variable importance, which with causal forests indicates how frequently each covariate was used for splitting, and saw that baseline SBP (0.29), baseline BMI (0.22), and age (0.13) were ranked highest.

3.2 Discussion

In this project, we used a causal forest to estimate the effect of tirzepatide versus liraglutide on one-year change in systolic blood pressure. Overall, we observed that tirzepatide was associated with a statistically significant SBP reduction relative to liraglutide. However, we were unable to find evidence of treatment-effect heterogeneity, as seen by our non-significant results in the differential forest prediction, best linear projection, and RATE. There are several plausible explanations for our results. One could be that our treatment effects are homogeneous across measured characteristics, which could imply that tirzepatide can be applied broadly without concern to specific subgroups over liraglutide if SBP reduction is a goal. Another potential explanation is that our analysis was not sufficiently powered to detect heterogeneity. While we had over 600 individuals in each treatment arm, as noted previously, the causal forest algorithm performs best with large sample sizes to account for honest splitting. Thus, we believe that replication in a larger cohort is necessary before concluding on the apparent absence of treatment-effect heterogeneity.

References

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Figures and Tables

Table 1: Baseline Characteristics of GLP-1 New Users by Ingredient

	Overall (N=1,287)	Liraglutide (n=681)	Tirzepatide (n=606)
Demographics			
Age, years, mean (SD)	55.5 (13.5)	56.4 (13.0)	54.5 (14.0)
Female, n (%)	778 (60.5)	394 (57.9)	384 (63.4)
Ethnicity, n (%)			
Hispanic	296 (23.0)	154 (22.6)	142 (23.4)
Non-Hispanic	958 (74.4)	512 (75.2)	446 (73.6)
Unknown	33 (2.6)	15 (2.2)	18 (3.0)
Race, n (%)			
White	812 (63.1)	408 (59.9)	404 (66.7)
Asian	112 (8.7)	65 (9.5)	47 (7.8)
Black	87 (6.8)	55 (8.1)	32 (5.3)
Other	276 (21.4)	153 (22.5)	123 (20.3)
Clinical Measures, mean (SD)			
Body mass index, kg/m ²	36.6 (8.3)	36.8 (8.8)	36.4 (7.7)
BMI category, n (%)			
Underweight	1 (0.1)	1 (0.1)	0 (0.0)
Normal	37 (2.9)	20 (2.9)	17 (2.8)
Overweight	206 (16.0)	117 (17.2)	89 (14.7)
Obese Class I	399 (31.0)	209 (30.7)	190 (31.4)
Obese Class II	295 (22.9)	136 (20.0)	159 (26.2)
Obese Class III	349 (27.1)	198 (29.1)	151 (24.9)
Body weight, lb	229.9 (57.6)	231.3 (61.7)	228.2 (52.5)
Systolic blood pressure, mmHg	130.9 (14.5)	130.9 (15.1)	130.9 (13.7)
Diastolic blood pressure, mmHg	76.3 (9.5)	75.2 (9.2)	77.6 (9.7)
Clinical Characteristics, n (%)			
Chronic kidney disease	195 (15.2)	115 (16.9)	80 (13.2)
Coronary artery disease	230 (17.9)	138 (20.3)	92 (15.2)
Diabetes	721 (56.0)	442 (64.9)	279 (46.0)
Heart failure	139 (10.8)	82 (12.0)	57 (9.4)
Hypertension	1080 (83.9)	589 (86.5)	491 (81.0)
Smoking status, n (%)			
Never	825 (64.1)	413 (60.6)	412 (68.0)
Former	356 (27.7)	214 (31.4)	142 (23.4)
Current	59 (4.6)	34 (5.0)	25 (4.1)
Unknown	47 (3.7)	20 (2.9)	27 (4.5)
Medication Burden, mean (SD)			
Antihypertensive medication classes	0.71 (1.03)	0.79 (1.06)	0.62 (0.99)
Proportion of days covered, 90-day	0.94 (0.06)	0.95 (0.06)	0.93 (0.06)

SD: standard deviation.

Table 2: Best Linear Projection of the Conditional Average Treatment Effect

Covariate	Estimate	Std. Error	t value	Pr(> t)
Intercept	1.848	8.834	0.21	0.834
Age	0.023	0.070	0.33	0.743
Female	-1.630	1.537	-1.06	0.289
Black	1.134	3.757	0.30	0.763
Other race	-2.724	3.255	-0.84	0.403
White	1.392	2.825	0.49	0.622
Non-Hispanic	-0.350	2.194	-0.16	0.873
Unknown ethnicity	-4.599	4.877	-0.94	0.346
Baseline BMI	0.117	0.106	1.10	0.271
Baseline SBP	-0.098	0.058	-1.70	0.090
Antihypertensive med. classes	-0.355	0.888	-0.40	0.690
Former smoker	-1.435	1.788	-0.80	0.422
Current smoker	3.954	3.477	1.14	0.256
Smoking unknown	-0.936	4.826	-0.19	0.846
Diabetes	2.105	1.728	1.22	0.223
Chronic kidney disease	2.116	2.501	0.85	0.398
Coronary artery disease	4.110	2.445	1.69	0.093
Heart failure	-1.774	3.056	-0.58	0.562

Uses heteroskedasticity-consistent (HC) 3 standard errors

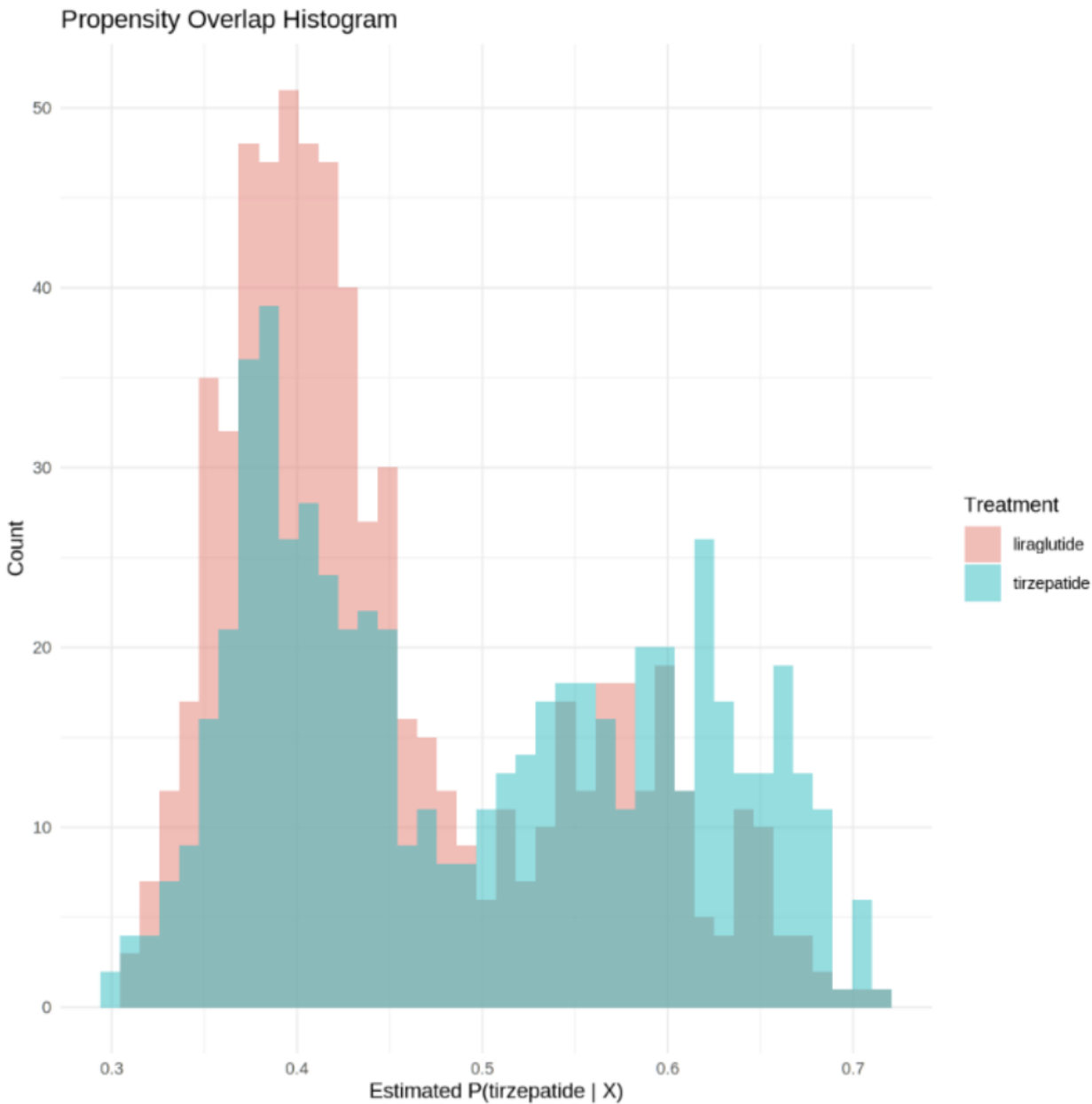


Figure 1: Propensity Overlap Histogram